20EC916 CAPSTONE PROJECT

HATE SPEECH DETECTION

USING ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

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PROJECT OVERVIEW

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WHAT IS

HATE SPEECH CLASSIFICATION

Hate speech classification is the ***process of identifying and categorizing speech*** *or language that may be considered offensive, harmful or prompting violence towards individuals or groups based on their race, gender, religion, nationality, or any other personal characteristics. This task is usually* ***performed by machine learning models*** *that have been trained on annotated data sets to recognize patterns in language and determine the level of hate speech present in a given piece of text.*

ABSTRACT

*A hate speech detection system **automatically identifies and flags hateful, abusive, or offensive language** in digital content, helping combat online toxicity and supporting content moderation on platforms like social media and forums. **Using machine learning and NLP techniques**, it analyzes text for linguistic patterns of hate speech. The process includes **data preprocessing (tokenization, stemming, stop-word removal)** and **model training** on labeled datasets. **Algorithms** like **SVM, Naive Bayes, RNNs**, and transformers (e.g., BERT) classify content effectively. Evaluation metrics (accuracy, precision, recall, F1) gauge performance, resulting in a **highly accurate system** for safe online environments and regulatory compliance.*

EXISTING SYSTEM

Current Approaches to Hate Speech Detection

- *Social Media Platform Models*
  - *Facebook's DeepText*
  - *Twitter's Text Classification Model*
  - *YouTube's Content ID System and Hate Speech Detection*
- *Open-Source Hate Speech Detection Models*
  - *HateSonar*
  - *HateXplain*
  - *Perspective API by Jigsaw (Google)*
- *Academic Models and Research Tools*
  - *Fused Neural Network*
  - *BERT-Based Models:*
- *Hybrid Approaches*
  - *Hate Speech and Offensive Language Detection (HASOC)*

LITERATURE SURVEY

MULTI-MODAL HATE SPEECH DETECTION USING MACHINE LEARNING

Challenge in Hate Speech Detection

- *Rapid growth in internet users and multimedia content.*
- *Difficulties in detecting hate speech in audio/video due to variations in tone, context (e.g., humor), and actions.*

Limitations of Single-Modality Detection

- *Existing models mostly use only text, which limits accuracy in complex media.*

Goal

- *Enhance accuracy in identifying hate speech across multimedia contexts.*

Published in: *2021 IEEE International Conference on Big Data (Big Data)*
Authors: *Fariha Tahosin Boishakhi, Ponkoj Chandra Shill, Md. Golam Rabiul Alam*

USAGE OF USER HATE SPEECH INDEX FOR IMPROVING HATE SPEECH DETECTION IN TWITTER POSTS

Challenges

- *Difficult to accurately detect hate speech in social media due to variations in context, user intent, and language.*
- *Handling large volumes of unstructured data from platforms like Twitter.*

Limitations

- *Single-modality models (e.g., text-only) may miss nuanced hate speech.*
- *Balancing data classes and managing outliers for accurate model performance.*

Goal

- *Develop a multi-faceted algorithm that calculates a user hate speech index to improve detection accuracy and effectively identify harmful content.*

Published in*: 2022 XXVIII International Conference on Information, Communication and Automation Technologies (ICAT)*

Authors*: Ehlimana Krupalija, Dženana Đonko, Haris Šupić*

BANGLA HATE SPEECH DETECTION SYSTEM USING TRANSFORMER BASED NLP AND DEEP LEARNING TECHNIQUES

Challenges

- *Difficulty in detecting hate speech in Bangla due to limited labeled data.*
- *Complex nature of Bangla text, requiring accurate translation and text preprocessing.*

Limitations

- *Dependence on translation (Bangla to English and back) may impact accuracy.*
- *Preprocessing challenges with emojis and regional linguistic nuances.*

Goal

- *Develop an automatic Bangla hate speech detection system using NLP and deep learning, aiming to enhance online safety in Bangla-language communities.*

Published in: *2023 3rd Asian Conference on Innovation in Technology (ASIANCON)*
Authors: *Omar Faruqe, Mubassir Jahan, Md. Faisal, Md. Shahidul Islam, Riasat Khan*

A TRUNCATED SVD FRAMEWORK FOR ONLINE HATE SPEECH DETECTION ON THE ETHOS DATASET

Challenges

- *Rising hate content on social media poses risks to individuals and groups.*
- *Users exploit platforms for malicious intent, sometimes for financial gain.*

Limitations

- *Existing methods may not effectively detect subtle or disguised hate speech.*
- *Difficulty in handling diverse expressions and contexts of hate speech.*

Goal

- *Use truncated SVD with the ETHOS dataset to enhance hate content detection, outperforming baseline models across multiple algorithms (SVM, Logistic Regression, XGBoost, and Random Forest).*

Published in: *2023 4th International Conference on Innovative Trends in Information Technology (ICITIIT)*

Authors: *Anusha Chhabra, Dinesh Kumar Vishwakarma*

A SHORT REVIEW ON HATE SPEECH DETECTION : CHALLENGES TOWARDS DATASETS AND TECHNIQUES

Challenges

- *Difficult to accurately define and detect hate speech due to variations in language, context, and intent.*
- *Dataset limitations impact detection effectiveness.*

Limitations

- *Multi-label classification across diverse datasets complicates detection.*
- *Classical and deep learning classifiers each have performance trade-offs.*

Goal

- *Review and evaluate machine learning architectures for hate speech detection, identifying key areas where improvements are needed.*

Published in: *2023 World Symposium on Digital Intelligence for Systems and Machines (DISA)*

Authors: *Manohar Gowdru Shridhara, Viktor Pristaš, Albert Kotvytskiy, L'ubomír Antoni, Gabriel Semanišin*

ROBUST IMPLEMENTATION OF A

HATE SPEECH DETECTION SYSTEM

KEY FEATURES

1.Multi-Modal Analysis: *Processes and classifies text, audio, and video content.*

2.Custom Algorithms: *Utilizes traditional (Naive Bayes, SVM) and deep learning models (GRU, BERT, LSTM) for hate speech detection.*

3.Dataset Creation & Preprocessing: *Combines public datasets and custom-collected data with preprocessing (tokenization, stemming, stopword removal, feature extraction).*

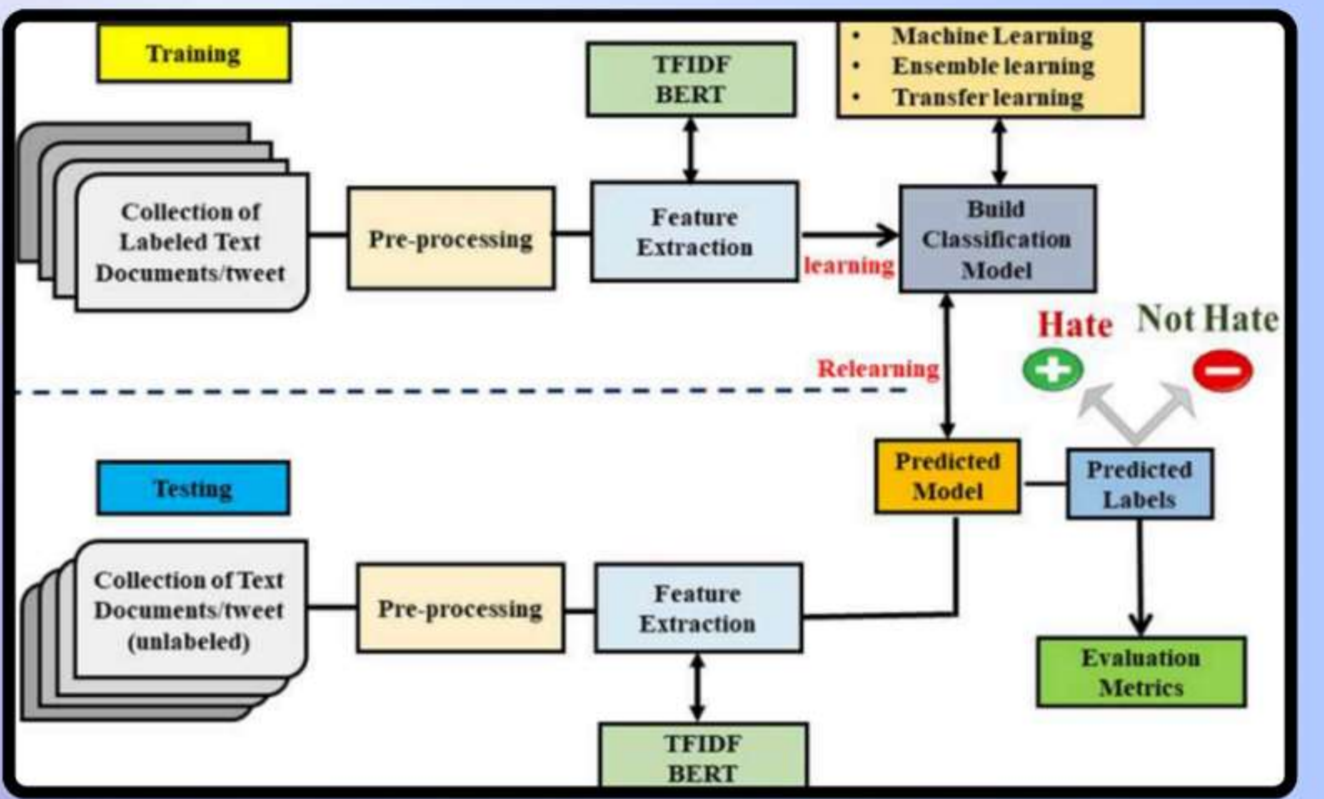
4.Accuracy Improvement: *Enhances performance via anomaly detection, data transformation, and model tuning.*

5.Real time Detection: *Provides immediate feedback/action (e.g., flagging, reporting) through a real-time detection pipeline.*

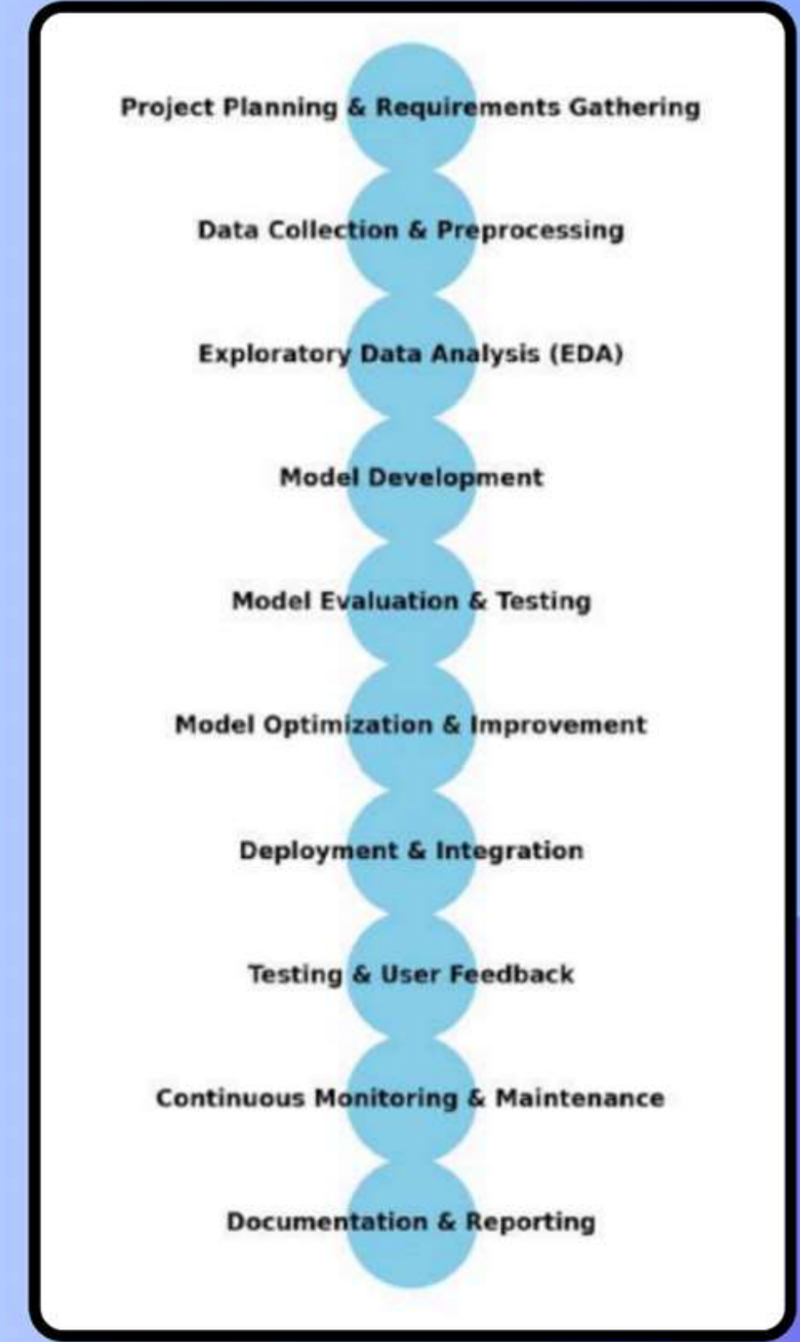
JUPYTER NOTEBOOK

1. *Interactive Environment*
2. *Support for Multiple Programming Languages*
3. *Rich Media Support*
4. *Easy to Share and Collaborate*
5. *Data Science and Machine Learning*
6. *Extensibility*
7. *Support for Machine Learning Frameworks*
8. *Ease of Use*
9. *Data Integration*

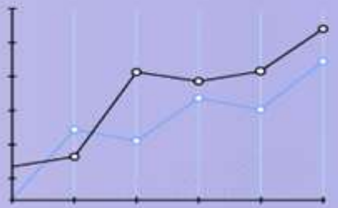
WORKFLOW



Training
TFIDF
BERT
• Machine Learning
• Ensemble learning
• Transfer learning
Collection of Labeled Text Documents/tweet
Pre-processing
Feature Extraction
learning
Build Classification Model
Hate
Not Hate
Relearning
Testing
Predicted Model
Predicted Labels
Collection of Text Documents/tweet (unlabeled)
Pre-processing
Feature Extraction
Evaluation Metrics
TFIDF
BERT



Project Planning & Requirements Gathering
Data Collection & Preprocessing
Exploratory Data Analysis (EDA)
Model Development
Model Evaluation & Testing
Model Optimization & Improvement
Deployment & Integration
Testing & User Feedback
Continuous Monitoring & Maintenance
Documentation & Reporting



PROJECT WORKING

In [8]:

```
test_data= "I will kill you"
df = cv.transform([test_data]).toarray()
print(clf.predict(df))
```

```
['Hate Speech detected']
```

In [9]:

```
test_data= "you are awesome"
df = cv.transform([test_data]).toarray()
print(clf.predict(df))
```

```
['No Hate and Offensive speech']
```

In [10]:

```
test_data= "you are bad i don't like you"
df = cv.transform([test_data]).toarray()
print(clf.predict(df))
```

```
['Offensive Language detected']
```

FUTURE SCOPE

- *Multimodal Hate Speech Detection*
- *Cross-Platform and Multilingual Support*
- *Real-Time Detection and Scalability*
- *Explainability and Transparency*
- *Improved User Profiling and Behavioral Analysis*
- *Ethical and Legal Implications*

In conclusion, this project lays the groundwork for developing a comprehensive, adaptable, and ethical hate speech detection system that can evolve with advancements in machine learning, natural language processing, and multimodal data integration. Future efforts can build on this work to create more inclusive, scalable, and transparent solutions to combat hate speech online.

THANK YOU!